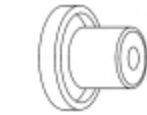


Valve Cover — LH

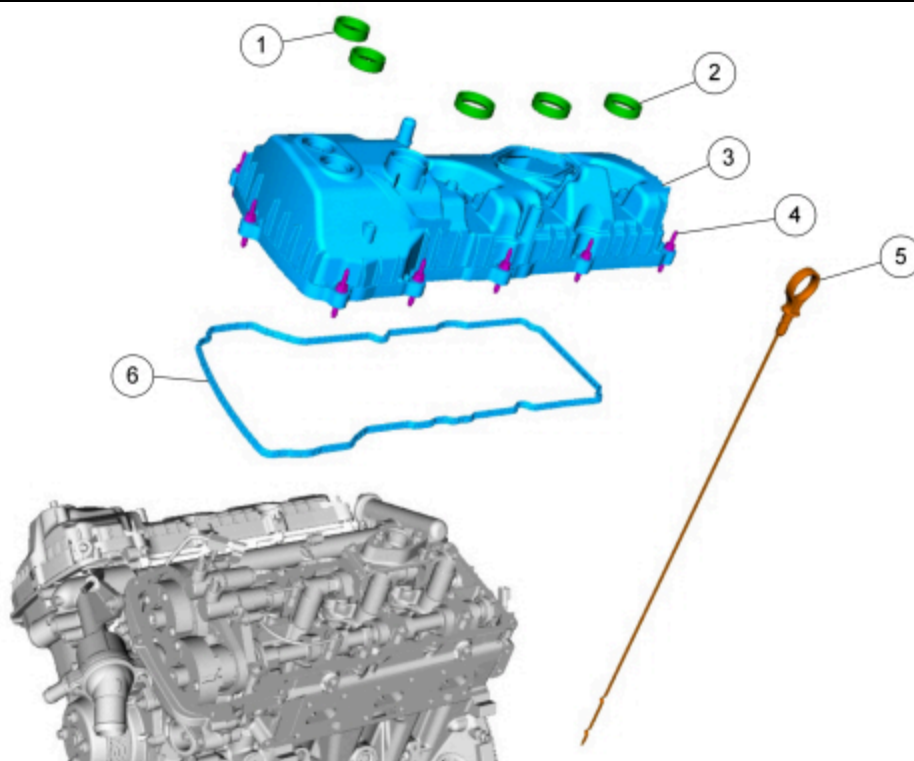
 [Ford Ecat Electronic Parts Catalog](#)

Special Tool(s)

 ST1326-A	Handle 205-153 (T80T-4000-W)
 ST2212-A	Installer, Differential Bearing Cone 205-142 (T80T-4000-J)
 ST2983-A	Installer, Spark Plug Tube Seal 303-1247/2
 ST2982-A	Remover, Spark Plug Tube Seal 303-1247/1

Material

Item	Specification
Motorcraft® High Performance Engine RTV Silicone TA-357	WSE-M4G323-A6
Motorcraft® Metal Surface Prep Wipes ZC-31-B	—
Motorcraft® Silicone Gasket Remover ZC-30-A	—

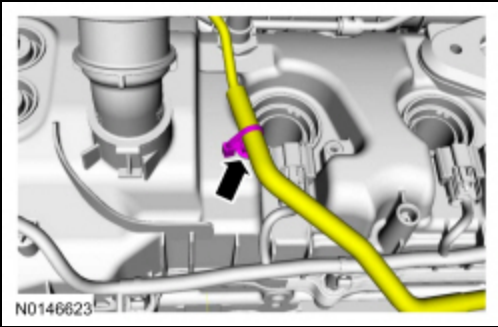


Item	Part Number	Description
1	6C535 6C535	Variable Camshaft Timing (VCT) solenoid seal (2 required)
2	6C535 6C535	Spark plug tube seal (3 required)
3	6582 6582	LH valve cover
4	00812 00812	Valve cover bolt and sleeve assembly (11 required)
5	6750 6750	Oil level indicator
6	6584 6584	LH valve cover gasket

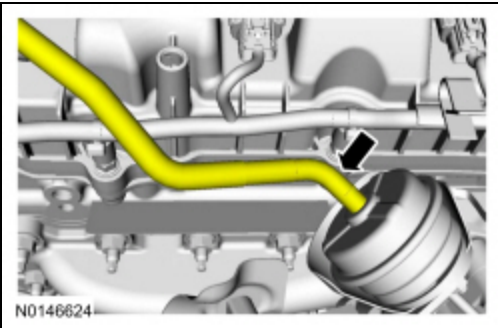
Removal

NOTICE: During engine repair procedures, cleanliness is extremely important. Any foreign material, including any material created while cleaning gasket surfaces that enters the oil passages, coolant passages or the oil pan, may cause engine failure.

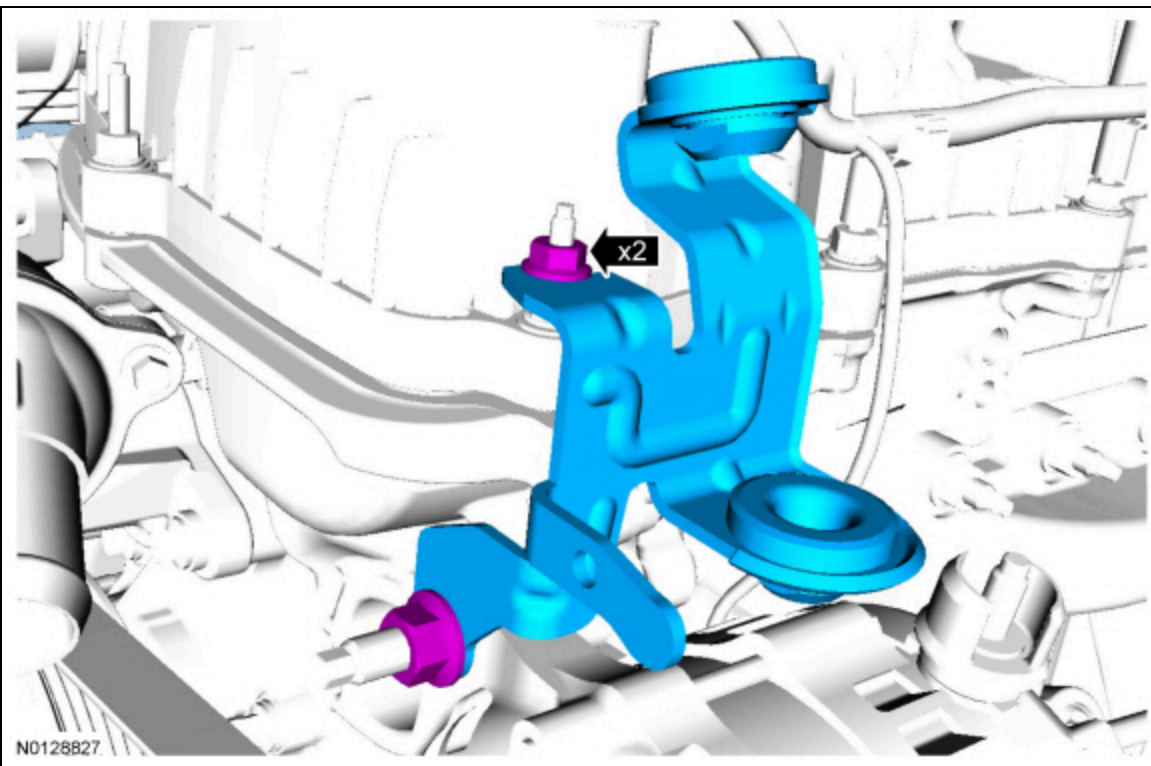
1. Remove the engine appearance cover.
2. Remove the Air Cleaner (ACL) outlet pipe and the LH ACL pipe-to-turbocharger pipe. Remove the LH turbocharger-to-Charge Air Cooler (CAC) tube. For additional information, refer to Intake Air System Components — Exploded View, 3.5L Gasoline Turbocharged Direct Injection (GTDI) in Section 303-12.
3. Remove the fuel injection pump. For additional information, refer to Section 303-04A.
4. Remove the ignition coil-over-plugs. for additional information, refer to Section 303-07A.
5. Remove the engine oil level indicator.
6. Remove the tie strap for the turbocharger wastegate hose.



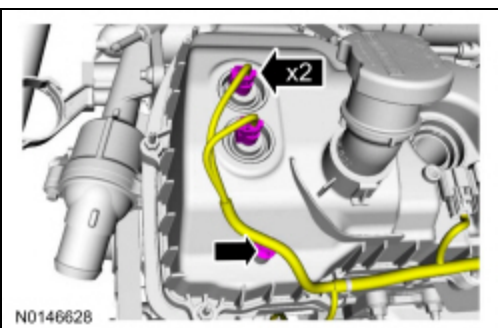
7. Disconnect the turbocharger wastegate hose and position aside.



8. Remove the 2 nuts and the CAC tube bracket.



9. Disconnect the 2 LH Variable Camshaft Timing (VCT) solenoid electrical connectors. Remove the tie strap.



10. Detach all of the wiring harness retainers from the LH valve cover and stud bolts.

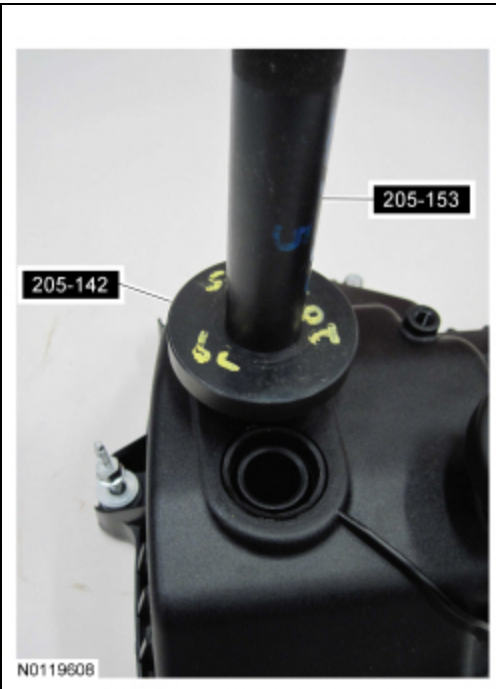
11. **NOTE:** When removing the valve cover do not apply excessive force to the VCT oil control solenoid. If the VCT solenoid seal sticks to the VCT oil control solenoid carefully wiggle the valve cover until the seal breaks free. After the valve cover is removed, inspect and replace the VCT solenoids if damaged. The plastic electrical connector on the VCT solenoid will normally rotate approximately 12 degrees inside the steel housing.

Loosen the 11 valve cover bolt and sleeve assemblies and remove the valve cover.

- Discard the valve cover gasket.

12. Inspect the 2 VCT solenoid seals. Remove any damaged seals.

- Using the Differential Bearing Cone Installer and Handle, remove the seal(s).



13. Inspect the spark plug tube seals. Remove any damaged seals.

- Using the VCT Spark Plug Tube Seal Remover and Handle, remove the seal(s).

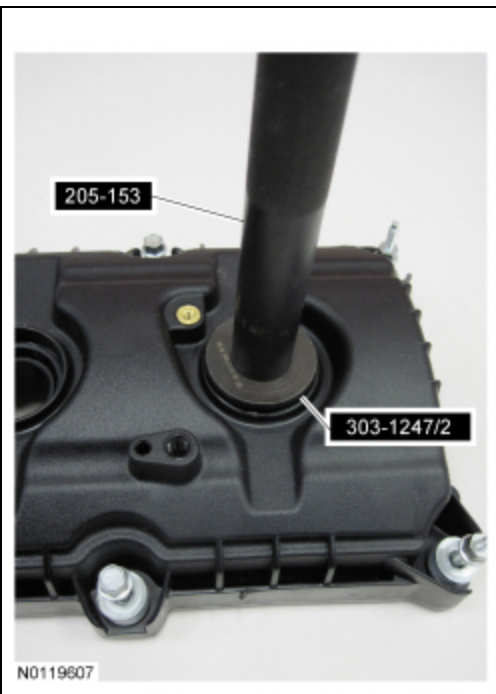


14. Clean the valve cover, cylinder head and engine front cover sealing surfaces with metal surface prep.

Installation

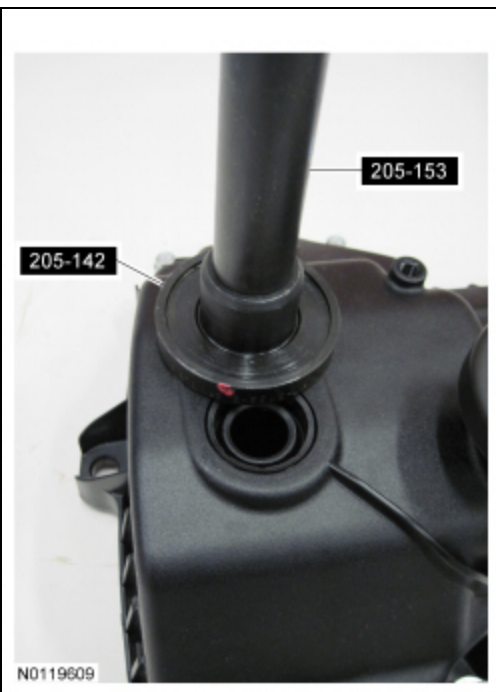
1. **NOTE:** *Installation of new seals is only required if damaged seals were removed.*

Using the **YCT** Spark Plug Tube Seal Installer and Handle, install new spark plug tube seals.



2. **NOTE:** *Installation of new seals is only required if damaged seals were removed.*

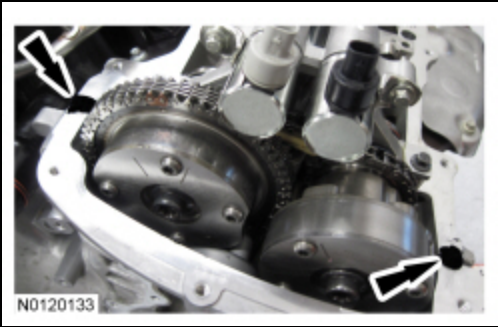
Using the Differential Bearing Cone Installer and Handle, install new **YCT** solenoid seal(s).



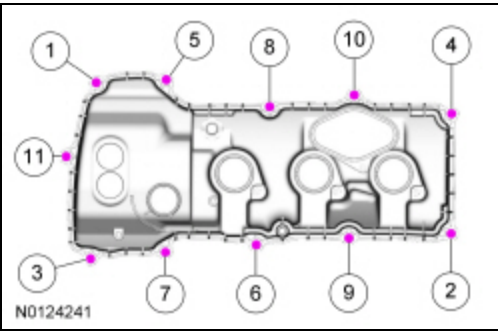
3. **NOTICE:** **Failure to use the correct Motorcraft® High Performance Engine RTV Silicone may cause the engine oil to foam excessively and result in serious engine damage.**

NOTE: *If the valve cover is not installed and the fasteners tightened within 4 minutes, the sealant must be removed and the sealing area cleaned. To clean the sealing area, use silicone gasket remover and metal surface prep. Failure to follow this procedure can cause future oil leakage.*

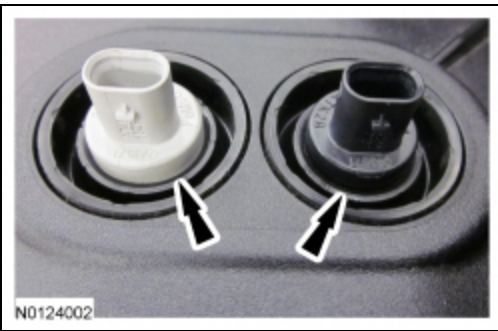
Apply an 8 mm (0.31 in) bead of Motorcraft® High Performance Engine RTV Silicone to the engine front cover-to-LH cylinder head joints.



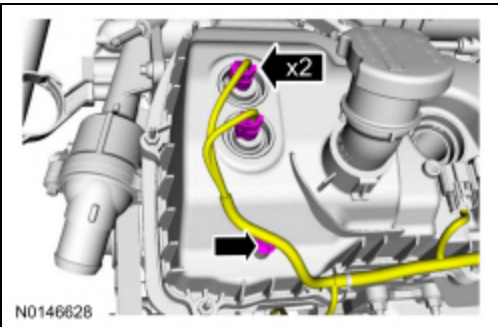
4. Using a new gasket, install the LH valve cover and tighten the 11 bolt and sleeve assemblies in the sequence shown.
 - Tighten to 10 Nm (89 lb-in).



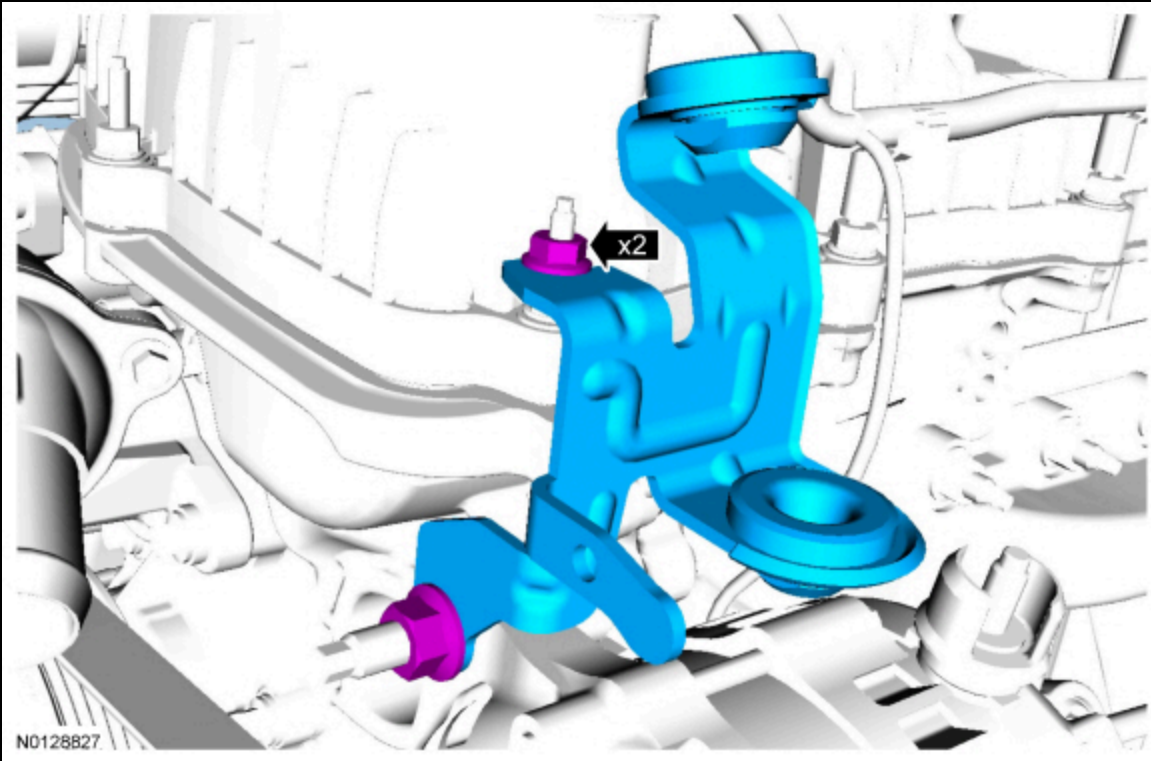
5. Make sure the VCT seals in the valve cover are below the top of the VCT oil control solenoid electrical connector or the VCT seal may leak oil.



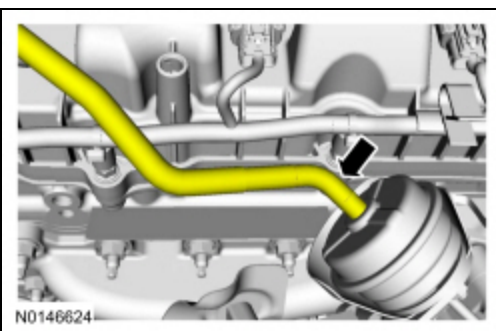
6. Attach all of the wiring harness retainers to the LH valve cover and stud bolts.
7. Connect the 2 LH VCT solenoid electrical connectors. Install a tie strap.



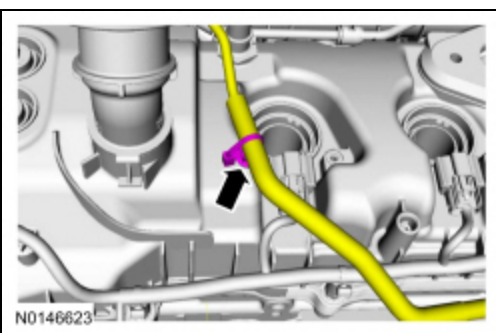
8. Install the CAC tube bracket and the 2 nuts.
 - Tighten the M6 nut to 6 Nm (53 lb-in).
 - Tighten the M10 nut to 48 Nm (35 lb-ft).



9. Connect the turbocharger wastegate hose.



10. Install the tie strap for the turbocharger wastegate hose.



11. Install the engine oil level indicator.

12. Install the ignition coil-over-plugs. for additional information, refer to Section 303-07A.

13. Install the fuel injection pump. For additional information, refer to Section 303-04A.

14. Install the LH turbocharger-to-CAC tube. Install the LH A/C pipe-to-turbocharger pipe and the A/C outlet pipe. For additional information, refer to Intake Air System Components — Exploded View, 3.5L GTDI in Section 303-12.

15. Install the engine appearance cover.